INCH-POUND

MIL-M-28787/292A
AMENDMENT 3
31 March 1993
SUPERSEDING
AMENDMENT 2
29 February 1984

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, UP/DOWN, BINARY, DIGITAL, KEY CODE BHA

This amendment forms a part of Military Specification MIL-M-28787/292A(EC), dated 24 June 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, 7, AND 8

*TABLE I: Delete and substitute new table I as printed on pages 3, 4, 5, 6, 7, 8, 9, 10, and 11 of this amendment.

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*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through HH shall be performed in any order with the following exceptions. Test conditions A, B, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, and F where applicable, followed by the remaining tests in the sequence."

*4.5.1, lines 5 and 6: Delete "D through EE, GG, and HH" and substitute "G through EE, GG, and HH".

*4.5.3, lines 3 and 4: Delete "D through EE, GG, and HH" and substitute "G through EE, GG, and HH".

*4.5.3, line 6: Delete "D through HH" and substitute "G through HH".

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6.4, second line: Delete "100°C" and substitute "85°C".

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*FIGURE 30, ITEM C01 box: Delete "ITEM C01" and substitute "ITEM C06".

*FIGURE 30, NOTE, first line: Delete "C01" and substitute "C06".

*FIGURE 30, NOTE, third line: Delete "Frequency = 120 Hz" and substitute "Frequency = 1 KHz".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0230)

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B18	VOUT (low-loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C18	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	μA
C19,C20	Input breakdown current	V	S	0	300	0	300	0	300	0	300	0	300	μA
D1-D18	Output short circuit current	VI	S	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
E1-E18	Output leakage current	VII	X	0	1	0	1	0	1	0	1	0	1	mA
F1-F20	Input clamp diode voltage	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G18	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H18	Output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V

- See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
J1-J53	Functional test	X	X	<—				See table X					—>	
K1-K18	Input current, low level	XI	X	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	-0.010	-0.435	mA
K19,K20	Input current, low level	XI	X	-0.03	-1.30	-0.03	-1.30	-0.03	-1.30	-0.03	-1.30	-0.03	-1.30	mA
L1-L18	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
L19,L20	Input current, high level	XI	X	0	60	0	60	0	60	0	60	0	60	µA
M1	Power supply current, high level	XII	X	10	102	10	102	10	102	10	102	10	102	mA
M2	Power supply current, low level	XII	X	10	102	10	102	10	102	10	102	10	102	mA
N1-N18	Transition time (tTLH), CL = 50 pF (carry, borrow and Q outputs)	XIII	S	5	50	5	50	5	55	5	50	5	55	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N19-N36	Transition time (tTLH), CL = 150 pF (carry, borrow and Q outputs)	XIII	Q	10	80	10	80	10	90	10	80	10	90	ns
P1-P18	Transition time (tTHL), CL = 50 pF (carry, borrow and Q outputs)	XIII	S	3	18	3	18	3	20	3	18	3	20	ns
P19-P36	Transition time (tTHL), CL = 150 pF (carry, borrow and Q outputs)	XIII	Q	6	40	6	40	6	42	6	40	6	42	ns
Q1-Q6	Propagation delay time (tPHL), CL = 50 pF (C/U to carry and C/D to borrow)	XIV	S	3	28	3	28	3	30	3	28	3	30	ns
Q7-Q12	Propagation delay time (tPHL), CL = 150 pF (C/U to carry and C/D to borrow)	XIV	Q	4	38	4	38	4	40	4	38	4	40	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R3	Propagation delay time (tPLH), CL = 50 pF (C/U to carry)	XIV	S	3	30	3	30	3	32	3	30	3	32	ns
R4-R6	Propagation delay time (tPLH), CL = 50 pF (C/D to borrow)	XIV	S	3	28	3	28	3	30	3	28	3	30	ns
R7-R9	Propagation delay time (tPLH), CL = 150 pF (C/U to carry)	XIV	Q	4	40	4	40	4	42	4	40	4	42	ns
R10-R12	Propagation delay time (tPLH), CL = 150 pF (C/D to borrow)	XIV	Q	4	38	4	38	4	40	4	38	4	40	ns
S1-S12	Propagation delay time (tPLH), CL = 50 pF (C/U to Q output)	XV	S	4	42	4	42	4	44	4	42	4	44	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S13-S24	Propagation delay time (tPLH), CL = 150 pF (C/U to Q output)	XV	Q	5	52	5	52	5	54	5	52	5	54	ns
T1-T12	Propagation delay time (tPHL), CL = 50 pF (C/U to Q output)	XV	S	5	51	5	51	5	53	5	51	5	53	ns
T13-T24	Propagation delay time (tPHL), CL = 150 pF (C/U to Q output)	XV	Q	6	61	6	61	6	63	6	61	6	63	ns
U1-U12	Propagation delay time (tPLH), CL = 50 pF (C/D to Q output)	XV	S	4	42	4	42	4	44	4	42	4	44	ns
U13-U24	Propagation delay time (tPLH), CL = 150 pF (C/D to Q output)	XV	Q	5	52	5	52	5	54	5	52	5	54	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
V1-V12	Propagation delay time (tPHL), CL = 50 pF (C/D to Q output)	XV	S	5	51	5	51	5	53	5	51	5	53	ns
V13-V24	Propagation delay time (tPHL), CL = 150 pF (C/D to Q output)	XV	Q	6	61	6	61	6	63	6	61	6	63	ns
W1-W12	Propagation delay time (tPLH), CL = 50 pF (load to Q output)	XVI	S	4	44	4	44	4	46	4	44	4	46	ns
W13-W24	Propagation delay time (tPLH), CL = 150 pF (load to Q output)	XVI	Q	5	54	5	54	5	56	5	54	5	56	ns
X1-X12	Propagation delay time (tPHL), CL = 50 pF (load to Q output)	XVI	S	4	44	4	44	4	46	4	44	4	46	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
X13-X24	Propagation delay time (tPHL), CL = 150 pF (load to Q output)	XVI	Q	5	54	5	54	5	56	5	54	5	56	ns
Y1-Y12	Propagation delay time (tPHL), CL = 50 pF (clear to Q output and minimum clear pulse width check)	XVI	S	3	39	3	39	3	41	3	39	3	41	ns
Y13-Y24	Propagation delay time (tPHL), CL = 150 pF (clear to Q output and minimum clear pulse width check)	XVI	Q	4	49	4	49	4	51	4	49	4	51	ns
Z1-Z6	Maximum counting frequency check (minimum count pulse width check)	XVII	S	844	948	844	948	844	948	844	948	844	948	ns

- See footnotes at end of table.

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						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA1-AA12	Setup time (data to load) and minimum load pulse width check	XVIII	S	110	130	110	130	110	130	110	130	110	130	ns
BB1-BB12	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
CC1-CC12	Output crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
CC13-CC24	Output crosstalk, duration	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
DD	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
EE	Power supply filter capacitance	XXI	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	μF
FF1-FF18	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF
FF19,FF20	Input capacitance	XXII	Q	4	30	4	30	4	30	4	30	4	30	pF
GG1-GG3	Borrow, pulse width	XXIII	Q	300	500	300	500	300	500	300	500	300	500	ns

- See footnotes at end of table.

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Test condition	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
HH1-HH24	Minimum load hold time	XXIV	S	170	230	170	230	170	230	170	230	170	230	ns

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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